

PUMP-204 CENTRIFUGAL PROCESS PUMP

Unit 2 - Cooling Water | Maintenance Department

1. Purpose

This synthetic demo manual describes daily inspection and basic care for PUMP-204. It is not a real plant document.

2. Machine data

- Tag: PUMP-204
- Type: horizontal centrifugal, single stage
- Service: cooling-water circulation
- Driver: 18.5 kW induction motor

3. Safety

Isolate electrical supply before opening the coupling guard.

Do not run the pump with the discharge valve fully closed for a long period.

4. What to inspect on PUMP-204

Complete this list during a walk-around inspection:

- Coupling guard in place and bolts tight
- Mechanical seal: no heavy leak, no steam, no burnt smell
- Bearing housings: feel for unusual heat; listen for growl or knock
- Baseplate and foundation bolts: no loose nuts, no cracked grout
- Suction and discharge valves: correct position, no leaked packing
- Pressure gauges: suction and discharge readings stable
- Motor fan cover: not blocked; no burnt insulation smell
- Oil level sight glass (if fitted): level between min and max

5. Abnormal sensor readings (demo limits)

Treat these as attention items, not an automatic trip:

- Vibration above 7.0 mm/s
- Bearing / casing temperature above 70 C
- Discharge pressure below 3.5 bar or above 6.5 bar
- Flow below 200 L/min
- Motor current above 28 A

If vibration and temperature rise together with falling flow, inspect the seal, bearings, and suction strainer first.

6. Daily checks

- Record vibration, temperature, pressure, flow, and motor current
- Look for new oil stains under the pump
- Confirm the inspection photo is taken from the drive-end seal area

7. Weekly checks

- Clean the strainer if flow is falling
- Recheck coupling alignment if vibration has stepped up
- Review the last maintenance history for seal or bearing work

8. Demo note

This file is pump_manual.pdf for the INDUSAI student prototype.

Related demo files: maintenance_history.pdf, inspection_procedure.pdf, inspection.jpg, sensor_data.csv.